

Digital Twin-Based Wind Load Assessment of a GDR Heritage Building under Climate Change

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Motivation:

Intensifying extreme wind events from climate change exceed design assumptions of GDR-era Plattenbau buildings (prefabricated reinforced-concrete panels, 1960–1990). Commercial SHM systems remain prohibitively expensive, proprietary, and black-box.

Research Objective:

This study develops low-cost, open-source MEMS accelerometers within a Digital Twin framework for wind load assessment of Weimar's Jakobsplan a GDR-era heritage building exceeding its design service life

Methodology & Results

Ambient vibration measurements were acquired at multiple floor levels. Recorded acceleration signals (a_x , a_y , a_z) were transformed to the frequency domain via Fast Fourier Transform (FFT) to identify the dominant natural frequencies of the structure.

FFT amplitude spectrum — acceleration channels a_x , a_y , a_z recorded at the Jakobsplan building. Dominant peaks indicate the principal natural frequencies of the structure.

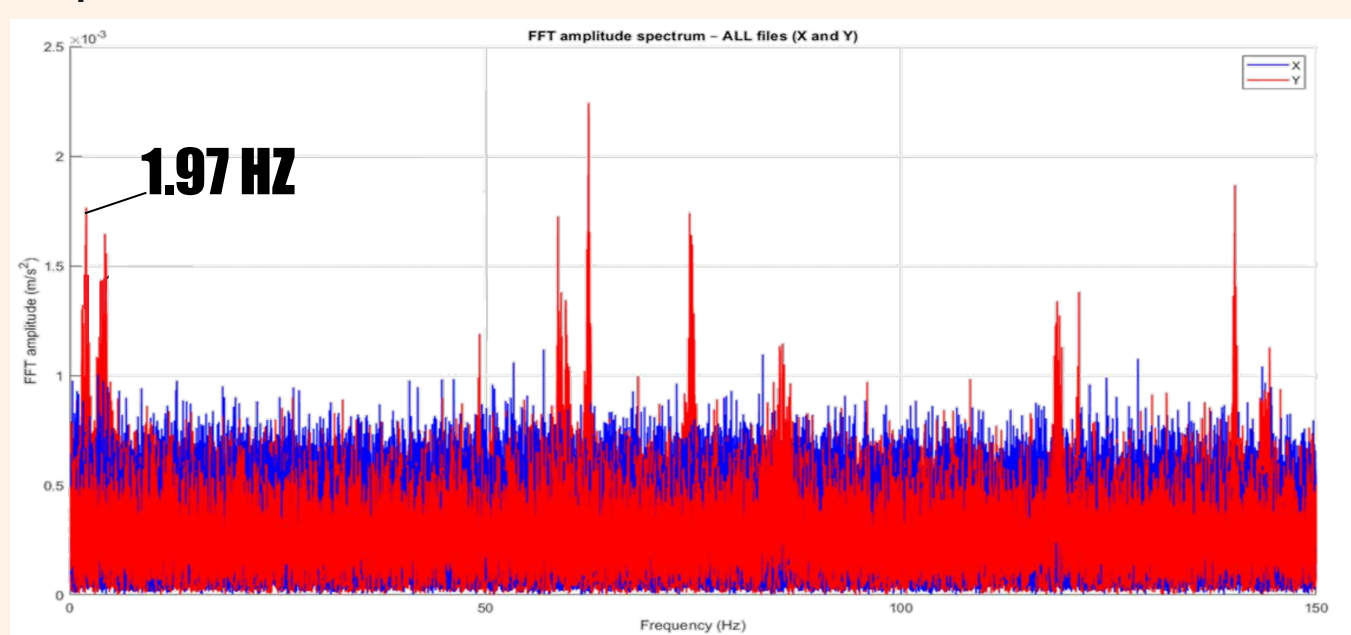
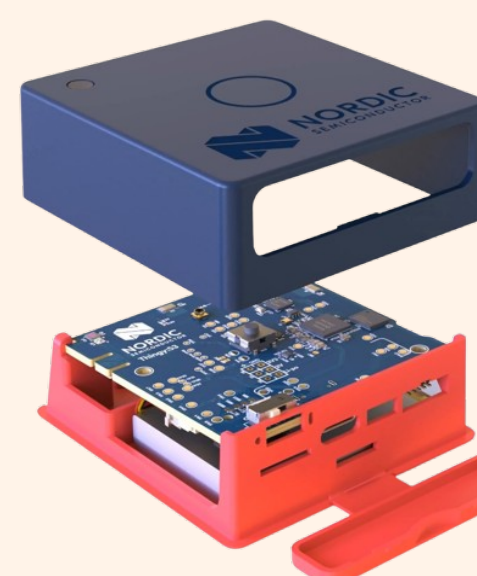
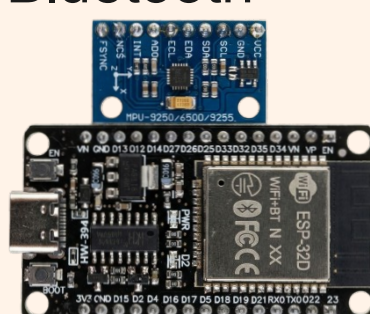


Figure 3: FFT Graph

Low-Cost IoT Sensor Platform

3 hardware platforms were evaluated for ambient vibration acquisition at ± 2 g full-scale range and 400 Hz sampling rate:

- Nordic Thingy:53 + ADXL362 — Bluetooth Low Energy 5.2
- Raspberry Pi Zero W + MPU-6050 — Wi-Fi / I²C
- ESP32 + MPU-6050 — Wi-Fi / Bluetooth



Bauhaus-Universität Weimar



Figure 1: Jakobsplan building, Weimar —
Exterior during construction (1971);
Exterior after renovation (2017)

Digital Twin Framework:

A four-component feedback loop connects the physical structure to its virtual replica: physical twin → data collection (IoT sensors) → digital entity (BIM, FEM, ML) → services and decision-making.

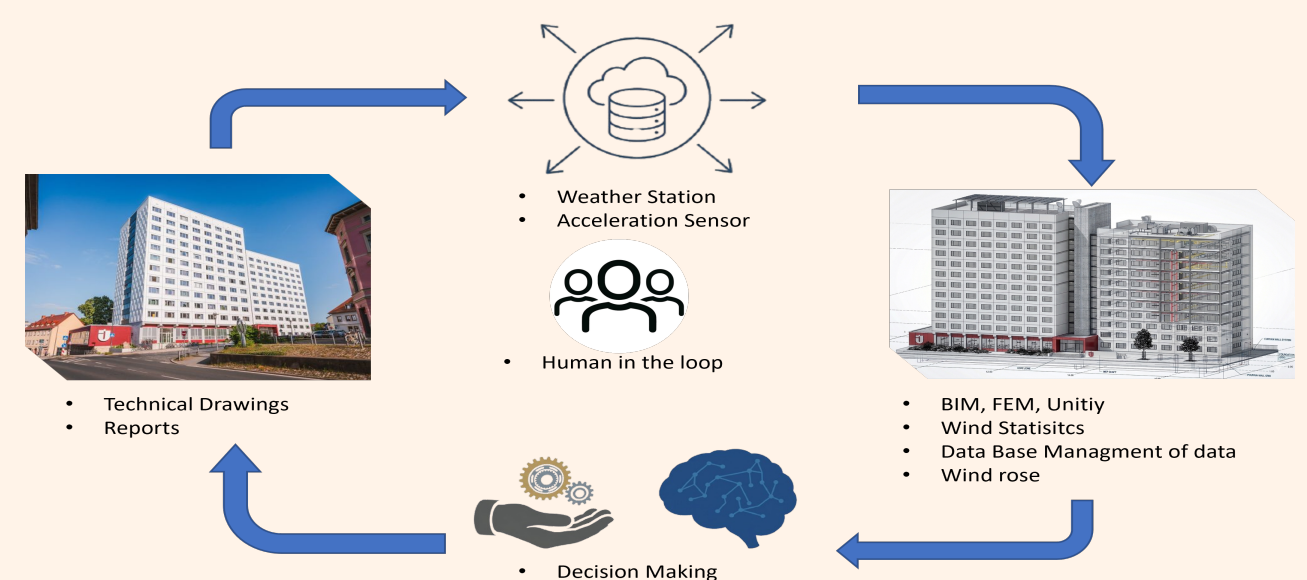


Figure 2: Digital Twin

Conclusions:

- Custom MEMS sensors identify the Jakobsplan dominant frequency at 1.97 Hz OMA-grade quality at low cost.
- MQTT PostgreSQL Unity pipeline delivers real-time acquisition and 3D visualization.
- Replicable open-source SHM blueprint for GDR-era buildings under climate change.